

Homework 6

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Section 003
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Collaboration Statement

I attended collaborative office hours and discussed Problem 6 with Professor Grace Work. Her guidance helped me better understand the structure of the proof and how to apply the Squeeze Lemma. All written work and final write-up are my own.

Problem 1 (Ross Chapter 2, 9.3)

Suppose $\lim a_n = a$, $\lim b_n = b$, and $s_n = \frac{a_n^3 + 4a_n}{b_n^2 + 1}$. Prove that $\lim s_n = \frac{a^3 + 4a}{b^2 + 1}$ carefully, using the limit theorems.

Discussion.

Here we essentially want to break the problem up into the individual limit components and apply the limit theorems 9.1-9.7 accordingly to show that the limit converges to $\frac{a^3 + 4a}{b^2 + 1}$.

For reference, here is what each of the 9.1-9.7 theorems say in a nutshell:

9.1 Theorem: Convergent sequences are bounded.

9.2 Theorem: If $\lim s_n = s$, then $\lim(ks_n) = k \lim s_n$.

9.3 Theorem: If $\lim s_n = s$ and $\lim t_n = t$, then $\lim(s_n + t_n) = \lim s_n + \lim t_n$.

9.4 Theorem: If $\lim s_n = s$ and $\lim t_n = t$, then $\lim(s_n t_n) = (\lim s_n)(\lim t_n)$.

9.5 Theorem: If $\lim s_n = s$, ($s \neq 0$) and $s_n \neq 0$ for all n , then $\lim \frac{1}{s_n} = \frac{1}{s}$.

9.6 Theorem: If $\lim s_n = s$, ($s \neq 0$), $s_n \neq 0$ for all n , and $\lim t_n = t$, then $\lim \frac{t_n}{s_n} = \frac{t}{s}$.

9.7 Theorem: (a) $\lim \frac{1}{n^p} = 0$ for $p > 0$.

(b) $\lim a^n = 0$ if $|a| < 1$.

(c) $\lim(n^{\frac{1}{n}}) = 1$.

(d) $\lim(a^{\frac{1}{n}}) = 1$ for $a > 0$.

Proof.

Given that $\lim a_n = a$ and $\lim b_n = b$, we can then express $\lim \frac{a_n^3 + 4a_n}{b_n^2 + 1}$ as $\frac{\lim a_n^3 + \lim 4a_n}{\lim b_n^2 + \lim 1}$. We can express the result of $\lim a_n^3$ as a^3 , and $\lim b_n^2 = b^2$ by theorem 9.4. Next we can show that $\lim 4a_n = 4a$ by theorem 9.2. Then, using the result from theorem 9.3, we can show that $\lim(a_n^3 + 4a_n) = a^3 + 4a$ and $\lim(b_n^2 + 1) = b^2 + 1$. Finally, using the result from theorem 9.6, we show that:

$$\lim s_n = \frac{\lim(a_n^3 + 4a_n)}{\lim(b_n^2 + 1)} = \frac{a^3 + 4a}{b^2 + 1}$$

for $b^2 + 1 \neq 0$ for all n . □

Problem 2 (*Ross Chapter 2, 9.9(b)*)

Suppose there exists N_0 such that $s_n \leq t_n$ for all $n > N_0$. Prove that if $\lim t_n = -\infty$, then $\lim s_n = -\infty$.

Discussion.

Here we want to leverage the definition of divergence which is given in Ross 9.8 theorem, which states that:

We write $\lim s_n = -\infty$ provided for each $M < 0$ there is a number such that $n > N$ implies $s_n < M$.

The nuance here is that we are comparing a sequence diverging to a 'negative infinity' (t_n) to another sequence that we are given to suppose is 'more negative' (s_n) after a certain point (N_0 ; for all $n > N_0$). In other words, we are comparing a sequence diverging to negative infinity (t_n) with another sequence (s_n) that is eventually less than or equal to it.

So if $\lim t_n = -\infty$ is true, then we know by theorem 9.8 that for every $M < 0$ there is a number N_1 such that $n > N_1$ implies $t_n < M$. We also know that there has got to be a N_0 (we suppose there is) such that $s_n \leq t_n$ for all $n > N_0$, so we can then just define an N that is greatest of the set of $\{N_0, N_1\}$ in whatever arbitrary case we take on (which we know much exist) so that we can say $s_n \leq t_n \leq M$ so that we can henceforth conclude that $s_n < M$ which is what we need to say that s_n diverges.

Proof.

Let $M < 0$. We know that there exists an N_1 such that $n > N_1$ implies $t_n \leq M$.

Let $N = \max\{N_0, N_1\}$. Thus, $n > N$ implies

$$s_n \leq t_n \leq M$$

which implies

$$s_n \leq M$$

Therefore $\lim s_n = -\infty$ □

Problem 3 (*Ross Chapter 2, 9.10(b)*)

Show that $\lim s_n = +\infty$ if and only if $\lim(-s_n) = -\infty$.

Discussion. We want to utilize theorem 9.8 and use a distinct M according to each case (since this is an iff statement) and prove divergence accordingly.

We will begin by show it first in the forward ($\implies$) direction: "if $\lim s_n = +\infty$, then $\lim(-s_n) = -\infty$, wherein we would let $M < 0$ so that we can say $s_n > -M$ then manipulate the expression to arrive at the conclusion that if $\lim s_n = +\infty$ then $\lim(-s_n) = -\infty$.

Then we would show the backward ($\impliedby$) direction of the iff statement such that: "if $\lim(-s_n) = -\infty$ then $\lim s_n = +\infty$ ", and follow the same procedure but with $M > 0$ to show the divergence toward a positive ∞ result.

Proof.

($\implies$) Let $\lim s_n = -\infty$. Let $M < 0$. For each $M < 0$ there is a number N such that $n > N$ implies

$$s_n > -M$$

which implies

$$-s_n < M$$

Thus, $\lim(-s_n) = -\infty$

($\impliedby$) Let $\lim(-s_n) = -\infty$. Let $M > 0$, For each $M > 0$ there us a number N such that $n > N$ implies

$$(-s_n) > -M$$

which implies

$$s_n > M$$

Thus, $\lim(s_n) = \infty$ □

Problem 4 (Ross Chapter 2, 10.2)

Prove Theorem 10.2 for bounded decreasing sequences.

Discussion.

Theorem 10.2 states that "All bounded monotone sequences converge." In the textbook, this theorem is proven first for the case of a bounded increasing sequence, using the concept of the supremum (*sup*). To prove the theorem for a bounded decreasing sequence, we instead use the infimum (*inf*) as our central idea. Intuitively, the argument mirrors the increasing case: rather than the terms of the sequence being squeezed upward toward the supremum, here they are squeezed downward toward the infimum.

So, given a bounded decreasing sequence (s_n) , we recognize that since it is bounded below, there exists an inf S , say L , which acts as the "greatest lower bound" of the set $\{s_n : n \in \mathbb{N}\}$. The proof strategy is to show that s_n converges to this infimum L using the ε -definition of convergence, just as the increasing case shows convergence to the supremum. The main difference is in the direction of the inequalities: where the increasing case used $s_N > u - \varepsilon$,

we will use $s_N < L + \varepsilon$. This reflects how a decreasing sequence approaches its lower bound from above.

Proof.

Let (s_n) be a bounded decreasing sequence. Since (s_n) is bounded, the set $S = \{s_n : n \in \mathbb{N}\}$ is bounded below. By the Completeness Axiom, there exists an infimum $L = \inf S$.

Let $\varepsilon > 0$. Since $L + \varepsilon$ is not a lower bound for S , there exists some $N \in \mathbb{N}$ such that $s_N < L + \varepsilon$. Because the sequence is decreasing, for all $n > N$ we have

$$s_n \leq s_N < L + \varepsilon.$$

We also know that, by definition of the infimum, $L \leq s_n$ for all n . Combining these inequalities, we get

$$L \leq s_n < L + \varepsilon,$$

which implies that

$$|s_n - L| < \varepsilon.$$

Therefore, $\lim s_n = L = \inf S$.

□

Problem 5 (Ross Chapter 2, 10.5)

Prove Theorem 10.4(ii).

Discussion.

Theorem 10.4 (ii) states:

If (s_n) is an unbounded decreasing sequence, then $\lim s_n = -\infty$.

To prove this, we should recall the definitions of an *unbounded* and a *decreasing* sequence.

A sequence (s_n) is called **decreasing** if for every $n \in \mathbb{N}$, $s_{n+1} \leq s_n$. It is called **unbounded below** if for every real number $M < 0$, there exists an $N \in \mathbb{N}$ such that $s_N < M$.

Together, these properties describe a sequence whose terms continually decrease without bound. Our goal is to show that this precisely means the limit of (s_n) diverges to negative infinity.

The strategy of the proof parallels that of part (i) of Theorem 10.4, which treats the unbounded increasing case (where $\lim s_n = +\infty$). Instead of showing the sequence eventually exceeds any given positive number, we will show that for every negative number M , the sequence eventually falls below it. We will also use the monotonic property (that s_n is decreasing) to ensure that once the sequence passes below M , it stays there for all subsequent n .

Proof.

To show that if (s_n) is an unbounded decreasing sequence, then $\lim s_n = -\infty$, let $M < 0$.

Since the set $\{s_n : n \in \mathbb{N}\}$ is unbounded and it is bounded above by s_1 , it must be unbounded below. Hence, there exists some $N \in \mathbb{N}$ such that $s_N < M$.

Because (s_n) is decreasing, for all $n > N$ we have

$$s_n \leq s_N < M.$$

Thus, for every $M < 0$, there exists an N such that $n > N$ implies $s_n < M$.

By definition of divergence to negative infinity, we conclude that

$$\lim s_n = -\infty.$$

□

Problem 6 (Ross Chapter 2, 10.7)

Let S be a bounded nonempty subset of $\mathbb{R}$ such that $\sup S$ is not in S . Prove there is a sequence (s_n) of points in S such that $\lim s_n = \sup S$. See also Exercise 11.11.

Discussion.

I am feeling really slow right now, so I am going to take things very slowly for this one. So let's begin by restate what we know. We know:

- (1) S is bounded. (So $\sup S$ exists by the Completeness Axiom.)
 - (a) S being bounded means there exists some $M \in \mathbb{R}$ such that for all $s \in S$, $|s| \leq M$. So $[-M, M]$ is an interval that contains S .
 - (b) From the completeness axiom, we know every nonempty set of real numbers that is bounded above has a supremum in $\mathbb{R}$. In other words we know S has a supremum by the completeness axiom since it is bounded.
- (2) S is nonempty: $S \neq \emptyset$. (So there is at least one point in our sequence).
- (3) $S \subseteq \mathbb{R}$.
- (4) $\sup S \notin S$.

From these three points, we can then construct a sequence (s_n) in S that converges to $\sup S$. The thing is, we do not know what $\sup S$ is exactly (it's arbitrary), but we do know that it is the least upper bound of S .

So to keep things arbitrary, we can define a t such that $t = \sup S$ and since $S \subseteq \mathbb{R}$, we know $t \in \mathbb{R}$. We know t is the least upper bound of S and must indeed exist for any arbitrary set S by the completeness axiom, but not an element of S .

Now, to construct our sequence (s_n) in S that converges to t , we can use the definition of supremum. Since t is the least upper bound of S , for every $\varepsilon > 0$, there exists an element $s \in S$ such that $t - \varepsilon < s \leq t$. This means we can find elements of S that are arbitrarily close to t , without t itself being in S .

So we can define our sequence (s_n) by choosing $s_n \in S$ such that $t - \frac{1}{n} < s_n \leq t$ for each natural number n . This way, as n increases, s_n gets closer and closer to t .

So that set up alone intuitively takes us to the conclusion that $\lim s_n = t = \sup S$. So this is awesome because now we have a sequence defined in a way that converges to $\sup S$. Half the battle is won.

The other half of the battle involves us proving convergence and then formalize this into a proof. Given our set up, we can easily prove convergence via the squeeze lemma (which we proved in *Chapter 2 §8, Exercise 8.5(a)*) which states that if we have three sequences (a_n) , (b_n) , and (c_n) such that $a_n \leq b_n \leq c_n$ for all n beyond some index N , and if $\lim a_n = \lim c_n = L$, then $\lim b_n = L$ as well. So check this proof out below.

Shoutout Professor Work for helping me with this one. My bad that my brain was exceptionally smooth when I came into office hours.

Proof.

Let $S \subseteq \mathbb{R}$ be a bounded nonempty set such that $\sup S \notin S$. Let $t = \sup S$. Since t is the least upper bound of S , for every $\varepsilon > 0$, there exists an element $s \in S$ such that $t - \varepsilon < s \leq t$.

Define the sequence (s_n) by choosing $s_n \in S$ such that $t - \frac{1}{n} < s_n \leq t \forall n \in \mathbb{N}$. In other words, for each natural number n , we select an element s_n from S that lies within the interval $(t - \frac{1}{n}, t]$.

From the squeeze lemma, we can then say that since $t - \frac{1}{n} < s_n \leq t$, we can define two sequences (a_n) and (c_n) such that $a_n = t - \frac{1}{n}$ and $c_n = t$. We can then see that $a_n \leq s_n \leq c_n$ for all n . Now we can compute the limits of (a_n) and (c_n) :

$$\lim a_n = \lim(t - \frac{1}{n}) = t - 0 = t$$

and

$$\lim c_n = \lim t = t$$

By the squeeze lemma, since $\lim a_n = \lim c_n = t$, we have $\lim s_n = t = \sup S$. □